

Lab8_Q2: Airport Priority Gate Assignment System

You are asked to design an airport passenger gate allocation system that assigns passengers to gates **only after all passengers have arrived**, not at the moment of arrival.

Key Functional Requirements

1. Passenger Collection (No Immediate Assignment)

- When a passenger arrives, they must **only be recorded**, not assigned to any gate.
- Assignment occurs **once all arrivals are completed**, through a dedicated scheduling operation.

2. Priority-Based Gate Distribution

After all passengers are collected:

- The system must allocate seats strictly by class:
 - VIP passengers must always be assigned first.
 - Business passengers must be placed next.
 - Economy passengers are assigned only if capacity remains.

3. Capacity Overflow Rules

If the total number of passengers exceeds the combined gate capacity:

- Remove (drop) passengers based on class in the following strict order:
 - 1. Drop Economy first / If still overflowing, drop Business
- No sorting functions may be used; priority must be enforced by **class-based bucketing** or equivalent logic.

4. Final Allocation

- After removal (if needed), passengers must be distributed into gates **in order**:
 - Fill Gate 1 completely / Then Gate 2 / And so on

5. Output Requirements

Your system must:

- Display any dropped passengers due to overflow and their class.
- Print each gate and its successfully assigned passengers.

Classes to Implement

You must implement **at least** the following two classes:

Gate

Airport

Input	Output
<pre>airport.add_passenger(("Alice", "Economy")) airport.add_passenger(("Bob", "VIP")) airport.add_passenger(("Charlie", "Business")) airport.add_passenger(("Diana", "VIP")) airport.add_passenger(("Ethan", "Economy")) airport.add_passenger(("Frank", "Business")) airport.add_passenger(("Gina", "VIP")) airport.assign_after_all_added() airport.print_gates()</pre>	<pre>Dropped due to priority/capacity: Alice (Economy) Dropped due to priority/capacity: Ethan (Economy) Gate 1: [('Bob', 'VIP'), ('Diana', 'VIP'), ('Gina', 'VIP')] Gate 2: [('Charlie', 'Business'), ('Frank', 'Business')]</pre>